Testicular & Erectile Dysfunction

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Block: Endocrine/Urology/ Reproduction

Thread: Clinical Medicine

Conflicts of Interest: None

Disclaimer: "Male" in this chapter refers to individuals born with Y chromosome, penis, and testicles. It does not refer to the individuals' gender identity.

Learning Objectives

By exam time, you should be able to:

1. Define infertility and recognize the contribution of male factors.
2. Describe the medical history as well as medication and substance exposures that may impact male fertility.
3. Describe the roles of physical exam, semen analysis, endocrine evaluation, and genetic workup in male infertility.
4. Describe the medical and surgical managements of male infertility.
5. Explain the 5 major types/causes of erectile dysfunction.
6. Describe the evaluation and management of erectile dysfunction.

Key Words

Spermatogenesis, varicocele, oligozoospermia, azoospermia, assisted reproductive techniques, erectile dysfunction

Further Reading

Eisenberg ML et al. Male infertility. *Nature Reviews Disease Primers.* 2023;9(1):49

Capogrosso P et al. Erectile dysfunction: update on clinical management. *European Urology*. 2025;88(4):388-399

Study Questions

1. Initial workup for male infertility should **always** include all the following EXCEPT:
    a. A thorough physical examination, with particular emphasis on genitalia and secondary sex characteristics
    b. A review of patient medications with attention to drugs affecting spermatogenesis, erection, ejaculation, or testosterone production
    c. A review of partner fertility and risk factors for infertility, as well as sexual history to determine the timing of intercourse and any other barriers to fertility
    d. Generally two semen analyses
    e. Scrotal ultrasound to assess presence or absence of varicocele

2. Laboratory evaluation of male infertility can include all the following EXCEPT:
    a. Generally two semen analyses to assess sperm numbers and motility
    b. Advanced sperm function tests such as DNA fragmentation index (DFI)
    c. Measurement of anti-Mullerian hormone (AMH) level
    d. Karyotype and Y chromosomal microdeletion (YCMD) test in patients with nonobstructive azoospermia (NOA)
    e. Cystic fibrosis transmembrane conductance regulator (CFTR) mutation test for patients with unilateral or bilateral absence of vas deferens

3. The following match an etiology of male infertility and its corresponding treatment EXCEPT:
    a. Kallmann syndrome: GnRH replacement therapy or gonadotropin replacement therapy (recombinant FSH and human chorionic gonadotropin [hCG])
    b. Hyperprolactinemia: Dopamine agonist (bromocriptine or cabergoline), pituitary radiation, surgical resection of prolactinoma
    c. Genital tract infection: Specific antimicrobial therapy
    d. Retrograde ejaculation: Alpha blocker (e.g. tamsulosin, terazosin)

e. Erectile dysfunction: Sex therapy, oral PDE5 inhibitors, vacuum erection device (VED), intracavernosal injection (ICI), penile implant surgery

4. A 65-year-old widower suffers erectile dysfunction and wishes to engage in sexual relations with a new partner. His medical history includes well-controlled diabetes mellitus, as well as occasional exertional angina, for which he takes sublingual nitroglycerin along with his antihypertensive regimen. All the following are potential treatment options, EXCEPT:

    a) Sex therapy

    b) Oral PDE5 inhibitor (e.g. sildenafil, tadalafil)

    c) Intracavernosal injection (ICI)

    d) Vacuum erection device (VED)

    e) Penile implant surgery

5. Which of the following treatments for prostate cancer can cause or worsen erectile dysfunction?

    a) External beam radiation therapy (EBRT)

    b) Robotic assisted radical prostatectomy

    c) Androgen deprivation therapy (ADT)

    d) Prostate brachytherapy

    e) All of the above

Infertility: Definition

1. Primary infertility: Inability to achieve pregnancy after **12 months** of regular, unprotected intercourse **(6 months if female partner is aged ≥35 years)**
2. Secondary infertility: As above, but in the setting of previously having at least one successful pregnancy and live birth

Infertility: Prevalence

1. The average conception rate in fertile, cis-hetero couples is approximately 85% within one year. Therefore, **15% of couples are infertile.**
2. **50**% of infertility cases involve male factor
    a. Female infertility: 30%
    b. Male infertility: 30%
    c. Mixed infertility: 20%
    d. Unexplained infertility: 20%

Testicular Reproductive Physiology

1. Normal spermatogenesis is predicated upon a coordinated release of hormones from the **hypothalamic-pituitary-gonadal (HPG) axis**
2. **Gonadotropin releasing hormone (GnRH)** is released by the hypothalamus in a pulsatile pattern every 90 - 120 minutes
3. GnRH is secreted into the pituitary portal blood system to act upon the gonadotrophic cells in the anterior pituitary to synthesize **follicle stimulating hormone (FSH)** and **luteinizing hormone (LH)**
4. LH stimulates the **Leydig cells** of the testicular interstitium to produce testosterone, while FSH supports spermatogenesis along the seminiferous tubules by stimulating the **Sertoli cells**
    a. Clinical implication: In the setting of poor sperm parameters on semen analysis, elevated serum FSH and LH levels will indicate **primary testicular failure** as the cause of infertility
5. This normal pattern of hormonal stimulation is self-regulated by steroid and peptide negative feedback loops. At the hypothalamus, testosterone and its metabolite, **estradiol**, reduce GnRH secretion. At the pituitary, estradiol (metabolized from testosterone by **aromatase)** suppresses LH secretion, while **inhibin B**, produced by Sertoli cells, suppresses FSH secretion.

   a. Clinical implication: **Intratesticular testosterone (ITT)** levels are >80 times higher than serum testosterone, and ITT is critical for spermatogenesis. The majority of men who use exogenous testosterone or anabolic steroids are infertile because the exogenous testosterone and its metabolites suppress FSH and LH secretion from the pituitary, leading to cessation of ITT production within the testicles.

6. The process of sperm maturation (spermatogenesis) within the seminiferous tubule requires approximately **64 - 74 days**. Sperm transport through the epididymis and vas deferens require an additional 14 days.

   a. Clinical implication: Semen analysis to assess the outcome of lifestyle interventions (e.g. nicotine cessation), medical therapy (e.g. hormone supplementation, antibiotic therapy), or surgery (e.g. varicocelectomy) should be performed approximately **90 days later** to realize the true effects of treatment.

7. The ejaculatory reflex arises from sensory input from the dorsal nerve of the penis entering the spinal cord at S2 - S4. Contraction of the seminal vesicles and bladder neck closure **(emission)** are controlled by sympathetic nerves arising from **T11 – L2** via the hypogastric plexus. These organs are richly supplied with alpha-adrenergic receptors. Seminal emission results in deposition of semen in the proximal urethra. The pulsatile phase of **ejaculation** is directed by involuntary contractions of the peri-urethral muscles (bulbospongiosus, ischiocavernosus, and levator ani) innervated by somatic nerves arising from **S2 - S4** levels.

   a. Clinical implication: **Retrograde ejaculation** is an occasional cause of male infertility that may result from various causes such as diabetes (neuropathy), surgeries (e.g. transurethral resection of prostate [TURP]), or medications (e.g. tamsulosin). In some of these patients, normal antegrade (forward) ejaculation can be restored by **alpha-adrenergic agonists (e.g. pseudoephedrine)**.

Examples of Male Infertility Causes

1. Pre-testicular: Idiopathic hypogonadotropic hypogonadism, Kallman syndrome, pituitary disease or tumor (e.g. prolactinoma), pituitary radiation or surgery, anabolic steroid use, androgen deprivation therapy (ADT), feminizing hormone therapy
2. Testicular: Klinefelter's syndrome (47,XXY), varicocele, undescended testicle, orchitis, testicular torsion, testicular cancer, environmental exposures (e.g. bisphenol A, organophosphate pesticides, excess heat), testicular radiation, chemotherapy, orchiectomy, febrile illness

3. Post-testicular: STI, epididymitis, prostatitis, congenital absence of the vas deferens (cystic fibrosis), occlusion of vas deferens (due to vasectomy or injury from inguinal hernia repair), ejaculatory duct obstruction, prostate surgery (e.g. prostatectomy, transurethral resection of prostate), retroperitoneal surgery (damage to sympathetic nerve fibers), diabetic neuropathy
4. Behavioral: Inaccurate timing or technique of sexual intercourse, use of lubricants (many contain **nonoxynol-9**, which is a spermicide)
5. Sexual dysfunction: Erectile dysfunction, hypoactive sexual disorder, decreased libido, Peyronie's disease, hypospadias, delayed ejaculation

Varicocele

1. Most common diagnosis seen in infertile men, and also the **most common reversible cause of male infertility**
2. Present in 15% of the general population
    a. Most of these patients are fertile with excellent sperm parameters
3. Among infertile men, varicocele is identified in **40% of men with primary infertility and 80% of men with secondary infertility**
4. In some patients, varicocele can also cause **chronic testicular pain**
5. **85% of patients with varicocele are left sided only.** 15% of patients have bilateral or right sided varicoceles. This is because the left testicular vein drains into the left renal vein at a right angle, whereas the right testicular vein drains into the inferior vena cava at an oblique angle, resulting in higher hydrostatic pressure on the left side.
6. Proposed mechanisms for impact of varicocele on male fertility include increase in intra-scrotal temperature, reflux of adrenal catecholamines through the internal spermatic vein, and relative hypoxia within the internal spermatic vein.
7. Varicocele is a clinical diagnosis based on physical exam and is graded on a scale of 1 – 3.
    a. Grade 1: Only palpable with Valsalva maneuver
    b. Grade 2: Palpable with and without Valsalva maneuver
    c. Grade 3: Veins are visible through the skin (**"bag of worms"** appearance)

    Generally, the higher the grade of varicocele, the greater the impact on spermatogenesis (but also the more improvement can be expected following varicocelectomy).

8. Scrotal ultrasound is not needed for routine diagnosis of varicocele, because varicocele that is seen on ultrasound but not palpated on exam ("subclinical" varicocele) is generally considered to be clinically insignificant. However, in certain patients who pose difficult exams (e.g. due to body habitus, robust cremasteric reflex), ultrasound may be helpful.
    a. On scrotal ultrasound, varicocele is present if **pampiniform plexus vein diameter is >2.5 - 3.0-mm, and the veins demonstrate retrograde flow (venous reflux) with Valsalva maneuver.**
9. Treatments for symptomatic varicocele (varicocele causing infertility or testicular pain) include **surgical varicocelectomy** (performed by a urologist) and **percutaneous angioembolization** (performed by an interventional radiologist).
    a. Indications for treatment: Abnormal semen analysis, chronic testicular pain, testicular size discrepancy (most common indication in pediatric patients)

Office Evaluation of Male Infertility

1. A fertility history should include inquiry into the patient's childhood development and surgical procedures, infection history and medical history with attention to endocrinopathy or delayed puberty.
2. A comprehensive medication history should be obtained with attention to drugs affecting spermatogenesis, ejaculation, or testosterone production.
3. A review of female partner fertility and risk factors should be performed. If there are any concerns (e.g. irregular menses, advanced maternal age, recurrent pregnancy losses), she should be referred for evaluation by female infertility specialist (i.e. REI).
4. Physical examination should search for any stigmata of hypogonadism (e.g. decreased hair distribution, bone and muscle mass, and gynecoid pelvis). The body mass index should be assessed and a search made for any inguinal or scrotal scars suggesting prior surgery. The penis should be inspected for hypospadias or other abnormalities.
5. Scrotal examination includes the assessment of testicular volume and consistency, identification of varicocele, and confirming the presence of both vas deferens.
    a. Normal testicular volume: **15 – 20-cc**
    b. Normal testicular length: **3.5 – 5.0-cm**
6. Digital rectal exam may reveal prostatitis or ejaculatory duct obstruction (associated with dilatation of seminal vesicles).

Laboratory Evaluation of Male Infertility

1. **Semen analysis** is the cornerstone of male fertility evaluation. If semen analysis is normal, no additional testing is required. However, if the semen analysis is abnormal, a repeat semen analysis should be obtained at least 2 weeks later.
2. The World Health Organization (WHO) defines the following normal reference parameters for semen analysis. This is based on semen analyses on fertile men whose partners conceived within 12 months. The numbers represent 5 percentile cutoffs in these fertile men.
    a. Volume: ≥1.5-ml
    b. Sperm concentration: ≥15 million/ml
    c. Sperm motility: ≥40% (progressive motility: ≥32%)
    d. Sperm vitality: ≥58%
    e. Normal sperm morphology: ≥4%
    f. Total motile sperm count (TMSC): ≥40 million
        i. Most important parameter
        ii. TMSC = volume x concentration x % motility
3. **Oligozoospermia** (low sperm concentration), **azoospermia** (no sperm seen), **asthenozoospermia** (low sperm motility), and **teratozoospermia** (low percentage of sperm with normal morphology) are some of the abnormalities identifiable on semen analysis.
4. Advanced sperm function tests may be indicated in some cases. The most utilized test is **sperm DNA fragmentation index (DFI)**. This is an assessment of DNA integrity, with higher percentage indicating increasing number of sperm with DNA strand breaks. High levels of sperm DFI (>20 – 30%) are associated with infertility, increased risk of miscarriage, and decreased success rates with assisted reproductive technologies (ART). Other, less frequently used sperm function tests include the detection of anti-sperm antibodies and assays that simulate sperm-egg interaction (e.g. sperm-cervical mucus penetration test, acrosome reaction, and zona-free hamster egg-sperm penetration assay).
5. Azoospermia can be classified as **obstructive (OA) or nonobstructive (NOA)**. In OA, spermatogenesis is functioning normally within the testicles, but a physical blockage prevents transport of sperm into ejaculate. Examples of OA causes include prior vasectomy, congenital absence of vas deferens, and ejaculatory duct obstruction. In NOA, there is impaired spermatogenesis within the testicles themselves. Examples of NOA causes

include genetic mutations, prior chemotherapy/radiation, and anabolic steroid use.

   a. With OA, there is near 100% sperm retrieval rate (SRR) from the testicles (with a surgical procedure). With NOA, SRR depends on etiology but is generally 40 - 50%.

6. In the NOA patient, genetic testing should include **karyotype** and **Y chromosome micro-deletion** testing. Karyotype can identify patients with conditions such as Klinefelter syndrome (47,XXY) or Robertsonian translocations. Approximately 15% of NOA patients demonstrate a detectable micro-deletion of the Y chromosome. Determination of the various regions of deletion can offer a prognosis for treatment in the in-vitro fertilization setting.
7. Patients with non-palpable vas deferens (congenital unilateral or bilateral absence of the vas deferens) should undergo genetic testing for **cystic fibrosis transmembrane conductance regulator (CFTR)**, as CFTR mutation is the most common cause. In addition, patients should undergo **renal ultrasound** because absent vas is associated with renal anomalies (e.g. renal agenesis, horseshoe kidney) due to common embryologic origin.
8. Patients with oligozoospermia, azoospermia, or small testicles should undergo serum endocrine testing (FSH, LH, and testosterone).

Medical Management of Male Infertility

1. Kallmann's syndrome: This genetic mutation is characterized by anosmia and congenital failure of GnRH secretion by the hypothalamus, leading to deficient FSH, LH and testosterone levels (**hypogonadotropic hypogonadism**). Treatment can be accomplished by GnRH replacement therapy or, more commonly, by gonadotropin replacement therapy using injectable recombinant FSH and human chorionic gonadotropin (hCG, which is an analogue of LH). These agents are injected to induce spermatogenesis and restore normal serum testosterone.
2. Hyperprolactinemia: This condition can arise in the setting of hepatic dysfunction or pituitary adenoma (prolactinoma). Symptoms may include visual field defects due to compression of the optic nerve by the prolactinoma. Galactorrhea and erectile dysfunction may occur due to low testosterone. Treatment can restore normal serum prolactin and testosterone levels with the use of dopamine agonists like bromocriptine or cabergoline. Pituitary radiation or surgical resection of the prolactinoma may be necessary.
3. Genital tract infection: If sexually transmitted infections (STI) are identified, specific antimicrobial therapy is indicated. NSAID therapy may be helpful in the setting of nonbacterial seminal vesiculitis.

4. Ejaculatory dysfunction: Retrograde ejaculation or absent emission due to prior retroperitoneal surgery, pelvic surgery, transurethral resection of the bladder neck or prostate, diabetic neuropathy, etc. may be treated with medical therapy to induce a forward flowing ejaculatory response. Alpha-adrenergic agonists such as pseudoephedrine and/or imipramine (a tricyclic antidepressant) may accomplish effective treatment for ejaculatory dysfunction in some patients.
5. Erectile dysfunction: If an erectile cause of infertility is identified, oral PDE5 inhibitors (e.g. sildenafil, tadalafil) or intracavernosal injections (e.g. Bimix, Trimix) may normalize erectile function.

Surgical Management of Male Infertility

1. Varicocelectomy: Surgical ligation of dilated spermatic veins is performed, usually under magnification with an operating microscope. Alternatively, percutaneous embolization of the spermatic veins may be utilized, although this approach is associated with higher failure and recurrence rates. Approximately 70% of patients undergoing varicocelectomy or embolization will show improved semen analysis parameters postoperatively. Several studies also suggest that varicocelectomy improves chances of successful assisted reproduction techniques (ART) such as IVF.
2. Vasectomy reversal: Performed for men with prior vasectomy who wish to regain natural fertility. The old vasectomy site scar is excised, and the testicular-side vas is inspected to see if fluid is expressed from the lumen. If fluid is present, and sperm is identified in the fluid under a microscope, then the testicular-side vas is anastomosed to the abdominal-side vas (**vasovasostomy**). If no fluid is identified from the testicular-side vas, this suggests that it has scarred down, and the abdominal-side vas is anastomosed to an epididymal tubule (**epididymovasostomy**). Both of these procedures require the use of operating microscope and fine 9-0 and 10-0 sutures (thinner than hair). Success rate ranges from >90% (if bilateral vasovasostomy is performed) down to 60 – 70% (if bilateral epididymovasostomy is performed).
3. Transurethral resection of ejaculatory ducts (TURED): Patients with ejaculatory duct obstruction at the level of the prostate may be treated by transurethral resection of the ejaculatory ducts to restore patency of the distal genital tract. However, this can cause complications such as chronic, recurrent epididymo-orchitis from urine reflux.
4. Surgical sperm retrieval: In the setting of azoospermia, sperm can be obtained directly from the testis or epididymis. Multiple techniques are available, such as testicular sperm aspiration (TESA), microsurgical testicular sperm extraction (microTESE), percutaneous epididymal sperm

aspiration (PESA), and microsurgical epididymal sperm aspiration (MESA). Patients with NOA should always undergo microTESE for maximum chance of successful sperm retrieval, which is 40 – 50%.

Erectile Dysfunction: Definition

1. Inability to attain and/or maintain penile erection sufficient for satisfactory sexual performance

Erectile Dysfunction: Prevalence

1. It is estimated that up to 50 million American men suffer from ED.
2. The prevalence generally matches the patient's age. 40% of 40-year-old men, 60% of 60-year-old men, and 80% of 80-year-old men suffer from some degree of ED.

Erectile Dysfunction: Major Types

1. Vasculogenic
    a. Most prevalent form, and the most common cause in men aged >50 years
    b. Due to impaired arterial inflow into the penis **(arterial insufficiency)**, failure of veno-occlusive mechanism **(venous leak)**, or both
    c. Associated with cardiovascular disease, diabetes mellitus, hypertension, hyperlipidemia, and nicotine use
    d. Considered to be a **harbinger of future cardiac events**. Penile arteries have much smaller diameter than cardiac arteries, and symptoms from atherosclerosis manifest years earlier. As such, men with new onset vasculogenic ED should undergo cardiovascular evaluation.
2. Neurogenic
    a. Due to disruption of neural pathways involved in erection
    b. Causes are varied and include examples such as stroke, spinal cord injury, pelvic injury or surgery (e.g. radical prostatectomy, abdominoperineal resection), and diabetic neuropathy
    c. Older men who undergo radical prostatectomy for prostate cancer often end up with severe ED due to combination of vasculogenic and neurogenic causes

3. Psychogenic
    a. More common in younger men
    b. Characterized by normal penile blood flow and nerve function but impaired erection due to psychological factors
    c. Common causes include performance anxiety, depression, relationship conflict, stress, and psychiatric disorders
    d. If a man has issues with partnered sexual activity but has no issues achieving and maintaining an erection with masturbation, it is usually a sign of psychogenic ED. These patients also tend to have spontaneous nocturnal/morning erections.
4. Hormonal
    a. Hypogonadism
    b. Hyperprolactinemia
5. Drug-induced
    a. Androgen deprivation therapy (ADT) is used to treat men with advanced/metastatic prostate cancer. Due to ADT causing chemical castration (i.e. undetectable testosterone), many patients on ADT suffer from severe ED.
    b. More commonly used medications associated with ED include antihypertensives (especially beta blockers and thiazide diuretics), psychotropic drugs (including selective serotonin reuptake inhibitors, tricyclic antidepressants, antipsychotics, and benzodiazepines), 5-α-reductase inhibitors (finasteride, dutasteride), and statins.

Erectile Dysfunction: Clinic Evaluation

1. Screening questionnaire
    a. Validated questionnaires commonly used in clinical practice include the **International Index of Erectile Function (IIEF)** and its abridged version, the **Sexual Health Inventory for Men (SHIM)**
2. Focused neuro-vascular history eliciting relevant risk factors
3. Medication history
4. Psychological and relationship assessment
5. Physical exam to assess obesity, neuro-vascular stigmata and signs of hypogonadism

Erectile Dysfunction: Additional Evaluation

1. Serum testosterone, glucose/hemoglobin A1c, and serum lipids may be checked to assess for etiologies
2. Duplex Doppler ultrasound with intracavernosal vasodilator injection
    a. A medication that induces erection (e.g. Trimix) is injected into the corpus cavernosum
    b. Once erection is achieved, Duplex Doppler ultrasound is used to assess the **peak systolic velocity (PSV)** and **end-diastolic velocity (EDV)** of the cavernosal arteries. Abnormalities in these parameters are indicative of vasculogenic ED, with PSV <25 cm/sec indicating arterial insufficiency and EDV >5 cm/sec indicating venous leak.
    c. This is not commonly performed due to the invasive nature of the study. However, this may be useful in certain situations. For example, if a 35-year-old obese male with anxiety and hypertension presents with significant ED, penile Doppler ultrasound may help distinguish between vasculogenic ED and psychogenic ED.

Erectile Dysfunction: Treatment

1. Behavioral modification (e.g. weight loss, physical activity, nicotine cessation, stress reduction)
2. Sex therapy / couples therapy
3. Vacuum erection device (VED)
    a. Can be used together with penile constriction band ("cock ring") to help maintain the erection
4. Medication adjustment (e.g. changing class of antihypertensive or antidepressant)
5. Oral phospodiesterase-5 (PDE5) inhibitors (sildenafil, tadalafil, avanafil, vardenafil)
    a. Mechanism of action
        i. Inhibition of PDE5 → **increased penile cavernosal cGMP** → smooth muscle relaxation in corpus cavernosum vasculature → increased erection hardness and duration
    b. Tadalafil has a long half-life and can be taken daily at low dose. Other agents are short acting and are taken as needed prior to sexual activity.

c. Common side effects include headache, flushing, nasal congestion, and myalgia

d. **Contraindicated in patients who have nitrate prescriptions** (e.g. nitroglycerin for chest pain)

i. Concomitant use of PDE5 inhibitor and nitrate can cause severe hypotension.

6. Intracavernosal injection (ICI)

a. Direct injection of vasodilators into the penile corpus cavernosum → direct smooth muscle relaxation

b. Safe to prescribe to patients who use nitrates

c. Commonly used ICI formulations include

i. Alprostadil (synthetic prostaglandin E1)

1. Efficacy: ~70% for men who failed oral PDE5 inhibitors

ii. Bimix (papaverine + phentolamine)

1. Limited efficacy data

iii. Trimix (alprostadil + papaverine + phentolamine)

1. Most commonly prescribed

2. Efficacy: Up to 90% in men who failed oral PDE5 inhibitors

d. Ischemic priapism

i. Most feared side effect of ICI (1 – 4% risk)

ii. Painful erection persisting **≥4 hours,** characterized by rigid shaft but **soft glans**

iii. Diagnosis is based on ABG performed on cavernosal blood

1. **PO_2 <30 mmHg, PCO_2 >60 mmHg, pH <7.25** are diagnostic

iv. Urologic emergency that necessitates prompt evaluation in ER

1. Priapism persisting **>36 hours** → permanent, complete ED as well as significant and irreversible penile scarring and shortening

v. Initial treatment consists of bedside penile irrigation and aspiration with normal saline as well as **phenylephrine** injections

1. Phenylephrine directly stimulates alpha-1 adrenergic receptors in the smooth muscle of the corpora cavernosa → vasoconstriction, decreased arterial inflow, and increased venous outflow → detumescence

vi. Refractory cases require surgical intervention ("shunt" procedure)

7. Penile implant surgery
    a. 2 major types: Inflatable penile prosthesis (IPP) and malleable penile prosthesis (MPP)
    b. Essentially a cure for ED and allows for spontaneous penetrative intercourse; however, this is an irreversible procedure (i.e. impossible to remove the implant and go back to using other treatments such as ICI)
    c. Highest satisfaction among all ED treatments (>95% in patients and partners), and ~40,000 implant surgeries are performed annually in USA
    d. Infection of the implant occurs in 1 – 4% of first-time cases and 10% of revision/replacement cases, and requires explantation of all implant components in addition to antibiotic therapy

Study Question Answers

1. Initial workup for male infertility should always include all the following EXCEPT:
    a. A thorough physical examination, with particular emphasis on genitalia and secondary sex characteristics
    b. A review of patient medications with attention to drugs affecting spermatogenesis, erection, ejaculation, or testosterone production
    c. A review of partner fertility and risk factors for infertility, as well as sexual history to determine the timing of intercourse and any other barriers to fertility
    d. Generally two semen analyses
    e. Scrotal ultrasound to assess presence or absence of varicocele

    Answer: E

    Scrotal ultrasound is not indicated in the initial work up of male infertility. The diagnosis of varicocele is based primarily on palpability of the spermatic cord veins on physical exam. Subclinical varicoceles (spermatic cord veins that appear enlarged on ultrasound but are not palpable on exam) are considered insignificant. Occasionally, scrotal ultrasound may be warranted if assessment of varicocele is difficult on physical exam (e.g. patient body habitus, robust cremasteric reflex).

2. Laboratory evaluation of male infertility can include all the following except:
    a. Generally two semen analyses to assess sperm numbers and motility
    b. Advanced sperm function tests such as DNA fragmentation index (DFI)
    c. Measurement of anti-Mullerian hormone (AMH) level
    d. Karyotype and Y chromosomal microdeletion (YCMD) test in patients with nonobstructive azoospermia (NOA)
    e. Cystic fibrosis transmembrane conductance regulator (CFTR) mutation test for patients with unilateral or bilateral absence of vas deferens

    Answer: C

    AMH is a quantitative marker of ovarian reserve in female patients and is not used in evaluation of infertile men

3. The following match an etiology of male infertility and its corresponding treatment except:

    a. Kallmann syndrome: GnRH replacement therapy or gonadotropin replacement therapy (recombinant FSH and human chorionic gonadotropin [hCG])

    b. Hyperprolactinemia: Dopamine agonist (bromocriptine or cabergoline), pituitary radiation, surgical resection of prolactinoma

    c. Genital tract infection: Specific antimicrobial therapy

    d. Retrograde ejaculation: Alpha blocker (e.g. tamsulosin, terazosin)

    e. Erectile dysfunction: Sex therapy, oral PDE5 inhibitors, vacuum erection device (VED), intracavernosal injection (ICI), penile implant surgery

    Answer: D

    Pseudoephedrine is an alpha-adrenergic receptor agonist frequently used in the treatment of ejaculatory dysfunction. Activation of alpha receptors stimulates smooth muscle contraction in the vas deferens, seminal vesicles, and bladder neck, thereby promoting seminal emission and antegrade ejaculation in select patients. Alpha blockers such as tamsulosin have the opposite effect and are a frequent iatrogenic cause of ejaculatory dysfunction.

4. A 65-year-old widower suffers erectile dysfunction and wishes to engage in sexual relations with a new partner. His medical history includes well-controlled diabetes mellitus, as well as occasional exertional angina, for which he takes sublingual nitroglycerin along with his antihypertensive regimen. All the following are potential treatment options, except:

    a) Sex therapy

    b) Oral PDE5 inhibitor (e.g. sildenafil, tadalafil)

    c) Intracavernosal injection (ICI)

    d) Vacuum erection device (VED)

    e) Penile implant surgery

    Answer: B

    PDE5 inhibitors are contraindicated in patients with nitrate prescriptions, as their combination may cause severe hypotension.

5. Which of the following treatments for prostate cancer can cause or worsen erectile dysfunction?

    a) External beam radiation therapy (EBRT)

    b) Robotic assisted radical prostatectomy

    c) Androgen deprivation therapy (ADT)

    d) Prostate brachytherapy

    e) All of the above

    Answer: E

    Prostate radiation and surgery have high risk of neurovascular compromise to the cavernous nerves that mediate erection. ADT leads to anorchid levels of testosterone (i.e. chemical castration).